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Complete Java Classes

by Kunal Sir

Quick Overview to The JavaScript

❖ JavaScript Data-Types: -

Data-types specifies what kind of data is going to be store in variable/data-member

Although, JavaScript is dynamically type language. It means JavaScript variables don't have any fix data-type instead of they dynamically adapt the data type at runtime, according to the value assigned to them.

Java script variable can store values according to the below data-types.

▪ Primitive data-types: -

1. string
2. number
3. boolean
4. undefine
5. null
6. bigInt
7. symbol

▪ Non-primitive data-types: -

1. Object
2. Array



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➤ Primitive data-types: -

1. **String:** - string is a sequence of characters that represents a text value.

In JavaScript string can be declare using “ ”, ‘ ’ and ` `.

In JavaScript if it is needed to store character then also

String data-type is used, as JavaScript don't have any special data type for characters.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-
width, initial-scale=1.0">
  <title>Document</title>
  <script>
    let name="pratik"
    let instituteName='cjc'
    // "Reat at cjc"
    let course=`React at ${instituteName}`
    console.log(course)
  </script>
</head>
<body>

</body>
</html>
```



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2. **Number:** - Number data type can represent integral as well as decimal (floating point) values.
3. **Boolean:** - Boolean data type represents logical entities. And can have one of two values, either true or false.
4. **Undefine:** - In JavaScript whenever variable is declared and it is not initialized with any value, then it has undefine data type.
5. **Null:** - Whenever it is not known which value that variable is going to be store, we must assign null value to the variable instead of remains uninitialized.
6. **BigInt:** - Whenever it is required to store very large integer number then we use BigInt data type.
7. **Symbol:** - Symbol data type created the unique values.

Example: -

```
<script>
  let name="pratik"
  let instituteName='cjc'
  // "Reat at cjc"
  let course=`React at ${instituteName}`
```



```
console.log(course)

let id=101;
console.log(id);
let marks=56.47;
console.log(marks)
let accountnumber=8765435676545577766// not store exact value
console.log(accountnumber)

let flag=true;
console.log(flag)
let data;
console.log(data)
let data2=null;
console.log(data2)
let examCenterCode=BigInt(765431.56526725681347658146357136);
console.log(examCenterCode)

</script>
```

❖ Non-Primitive Data-Types: -

In JavaScript, Object and Array are the non-primitive datatypes.

1. **Object:** - JavaScript Object represents an instance from which we can access properties and their corresponding values.

Object are used to store many properties and their values in single reference.

▪ **Features of JavaScript Object:** -

- JavaScript objects can be define using object literals `{ }`.
- JavaScript objects are slightly similar to the java Hashmap.
- We can update the values of JS object properties after declaration.
- We can add new properties into the JS object after declaration.



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Example: -

```
<script>
//Define JS Object using object literals
let student={
    rollno:101,
    studentName:'abc',
    courseName:'java',
    batchNumber:'FDJ B-176',
    contactNumber:9876543210
}
// print object
console.log(student);           // {rollno: 101, studentName: 'abc', courseName: 'java',
                                //   batchNumber: 'FDJ B-176', contactNumber: 9876543210}

// Assess/print Object properties.
console.log("Student RollNo:- "+student.rollno);           // Student RollNo:- 101
console.log("Student Name:- "+student.studentName);        // Student Name:- abc
console.log("Course Name:- "+student.courseName);          // Course Name:- java
console.log("Batch Number:- "+student.batchNumber);        // Batch Number:- FDJ B-176
console.log("Contact Number:- "+student.contactNumber);    // Contact Number:- 9876543210
// Modify value of object property
student.studentName="pqr";
console.log(student);           // {rollno: 101, studentName: 'pqr', courseName: 'java',
                                //   batchNumber: 'FDJ B-176', contactNumber: 9876543210}
console.log("Student Name:- "+student.studentName);        // Student Name:- pqr
// Adding new Property
student.email="pqr@gmail.com";
console.log(student); // {rollno: 101, studentName: 'abc', courseName: 'java',
                        //   batchNumber: 'FDJ B-176', contactNumber: 9876543210,
                        //   email: 'pqr@gmail.com'}
console.log("Student Email:- "+student.email);             // Student Email:- pqr@gmail.com
</script>
```



2. Array: - In JavaScript, array is use to represent group of elements/entities.

Arrays are the special variables that can hold more than one value. We can access them using their index.

▪ Features of JavaScript Array: -

- a) JavaScript arrays don't have fix size, they can grow and shrink at runtime.
- b) We can define array using array literals `[ ]`.
- c) JavaScript array can have heterogenous elements.
- d) JavaScript array has length property to check number of elements stored in array.

```
//Define Array.  
  
//index no 0    1    2    3    4    5    6  
let days=['SUN','MON','TUE','WED','THU','FRI','SAT'];  
console.log("Array:- "+ days);    //Array:- SUN,MON,TUE,WED,THU,FRI,SAT  
  
//  
// Geting array legth(size)  
console.log(`Array contains ${days.length} elements`); //Array contains 7 elements  
// Assessing array elements;  
console.log("0th index:- "+days[0]);    // 0th index:- SUN  
console.log("1st index:- "+days[1]);    // 1st index:- MON  
console.log("2nd index:- "+days[2]);    // 2nd index:- TUE  
console.log("3rd index:- "+days[3]);    // 3rd index:- WED  
console.log("4th index:- "+days[4]);    // 4th index:- THU  
console.log("5th index:- "+days[5]);    // 5th index:- FRI  
console.log("6th index:- "+days[6]);    // 6th index:- SAT
```